SkyPredict

Smart Weather Monitoring and Prediction System Using Sensor Fusion and Machine Learning

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Contents

Introduction

Waste management stands as a critical challenge for nations worldwide, and Egypt is no exception. With growing concerns over environmental sustainability and economic development, the imperative to address waste management issues has never been more pressing. The Egypt Grand Challenge presents a unique opportunity to pioneer innovative solutions that not only recycle garbage and waste but also foster economic growth and environmental resilience. This research aims to explore the integration of advanced technologies, such as *the Internet of Things (IoT)*, to revolutionize waste management practices in Egypt. By leveraging a feedback system and other modern methodologies, this endeavor seeks to enhance the scientific and technological environment, mitigate the impacts of climate change, and bolster Egypt's industrial and agricultural sectors. In the next pages we outlined a multifaceted approach towards achieving these objectives, paving the way for a more sustainable and prosperous future for Egypt.

1

Present **and Justify a Problem and Solution Requirements**

Egypt's Grand Challenges

To start our project, we had to identify the grand challenges to be solved:

Improve the use of alternative energies.
Recycle garbage and waste for economic and environmental purposes.
Deal with urban congestion and its consequences.
Work to eradicate public health issues/diseases.
Increase the industrial and agricultural bases of Egypt.
Address and reduce pollution fouling our air, water, and soil.
Improve uses of arid areas.
Manage and increase the sources of clean water.
Deal with population growth and its consequences.
Improve the scientific and technological environment for all.
Reduce and adapt to the effect of climatic change.

Related Grand Challenges:

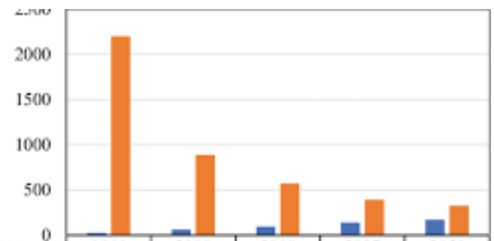
1-Managing and Increasing Clean Water Sources in Egypt

Water scarcity poses a critical challenge to Egypt's economic and social stability. The country heavily depends on the Nile River, which provides over 97% of its freshwater. However, rapid population growth, climate change, and upstream developments threaten this vital resource. Managing and increasing clean water sources is essential for sustaining Egypt’s agriculture, industry, and daily life. This section explores the scope of the water challenge, Egypt’s response, and potential technological solutions, including the role of machine learning in predicting rainfall and optimizing water resources.

The Water Challenge: Current State and Risks

Egypt's water resources are under significant stress, with the country classified as “water-scarce” by the United Nations. Egypt’s per capita water availability is 560 cubic meters per year as of 2024, well below the international water poverty line of 1,000 cubic meters per year. Factors contributing to this crisis include:

- **Climate Change**: Rising temperatures and changing precipitation patterns threaten water supply. By 2050, water demand is expected to increase by 20%, while supply may decrease by 15% due to environmental factors.
- **Upstream Developments**: The construction of the Grand Ethiopian Renaissance Dam (GERD) upstream has heightened concerns about reduced Nile water flow. If the dam’s operations aren’t managed cooperatively, Egypt could lose up to 22% of its water supply.
- **Agriculture and Industry:** Agriculture consumes over 80% of Egypt's freshwater, but inefficient irrigation practices result in significant water loss. Additionally, industrial growth increases pressure on water resources, exacerbating the scarcity.



year	1960	2000	2018	2037	2050
Population (million)	25	62	97	140	170
Nile Water Availability (m3/capita/year)	2200	887	570	392	324

Figure (1) shows the trend of declining per capita water availability in Egypt from 2010 to 2024 and the projected figures up to 2050.

Economic Impact of Water Scarcity

Water scarcity has severe repercussions for Egypt's economy. Without effective water management, Egypt's GDP could decrease by 1.5% to 2.5% annually by 2030 due to reduced agricultural productivity, higher water costs, and environmental damage. Agriculture, a sector that accounts for 11% of Egypt's GDP and employs nearly 25% of the labor force, is particularly vulnerable to water shortages.

- Agricultural Decline: A 10-15% reduction in agricultural output is projected by 2030 due to water scarcity, leading to higher food prices and increased import dependency.

- Industrial Losses: Water-intensive industries such as textiles and food processing face operational challenges, leading to reduced output and higher costs, potentially cutting 5% of industrial GDP by 2035.

Egypt's Initiatives to Increase Clean Water Sources and solutions Egypt Has Implemented

Egypt has already launched several projects aimed at addressing water scarcity, ensuring efficient water use, and securing alternative sources of clean water.

1. Desalination Projects

- Expanding Desalination Capacity: Egypt is investing heavily in desalination plants, with plans to construct 47 desalination facilities along the Mediterranean and Red Sea coasts by 2030. These plants aim to produce 2.44 million cubic meters of water per day by 2025.
- Economic Impact: Desalination is a high-cost solution, but if efficiently integrated, it could offset 20% of Egypt's water demand in coastal areas by 2035, reducing pressure on the Nile and providing water for industrial zones and urban centers.

2. Smart Water Management Systems

- Toshka Project: This large-scale water diversion and irrigation project aims to redistribute Nile water to desert areas, boosting agricultural land by 540,000 hectares. Utilizing smart irrigation systems, the project ensures optimal water use, reducing waste.
- Canal Lining and Rehabilitation: Egypt has begun a comprehensive canal lining project to reduce water loss through seepage, saving an estimated 5 billion cubic meters of water annually.
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Predicting Rainfall Through Machine Learning:

One of the most promising solutions for improving water management is using machine learning (ML) to predict rainfall and optimize water usage.

1. Machine Learning for Rainfall Prediction

- Precision Forecasting: Machine learning models can analyze vast datasets, including historical weather patterns, temperature trends, and satellite imagery, to predict rainfall more accurately. This allows for better water resource planning and agricultural decision-making.
- Economic Benefits: By predicting rainfall and ensuring efficient irrigation based on precise forecasts, Egypt can save up to 30% of water used in agriculture. This translates to significant cost savings and better crop yields, potentially increasing agricultural GDP by $3 billion annually by 2030.

2. Implementing ML in Water Management

- Irrigation Optimization: ML systems can be integrated into smart irrigation networks, using predictive analytics to determine the best times and amounts of water for crops. This approach is being tested in pilot projects in Egypt's desert farms, with early results showing water use reductions of up to 25%.

Existing Solutions:

- Wastewater Treatment and Reuse: Egypt has already established a strong wastewater treatment and reuse system, with plants like the Bahr El-Baqar wastewater treatment plant the largest in the world, producing 5.6 million cubic meters of treated water daily for agricultural use.
- National Water Resources Plan 2037: This strategy aims to improve water efficiency, focusing on desalination, wastewater reuse, and irrigation modernization to ensure water security for the future.

Securing Egypt's Future through Water Management

Addressing the water scarcity challenge is critical to Egypt's economic stability and growth. By combining smart water management systems, desalination, and advanced technologies like machine learning for rainfall prediction, Egypt can secure its water future. These efforts will boost agricultural productivity, support industrial growth, and protect vulnerable communities from climate-induced water shortages.

Implementing these solutions could increase Egypt's agricultural GDP by **$3-5 billion** by 2035 and safeguard its broader economy from the impacts of water scarcity. Through comprehensive, forward-looking water management policies, Egypt is poised to build a more resilient and prosperous future.

2-Improving the scientific and technological environment for all:

Improving Egypt's scientific and technological environment is crucial for fostering national growth, innovation, and development across various sectors. The country's current challenges span its education system, research and development (R&D), and academia-industry collaboration. Let's dive into some statistics that illustrate Egypt's position and challenges in the global scientific and technological landscape:

Key Statistics on Egypt's Scientific and Technological Environment:

1. Global Ranking in Science: According to the SCImago Journal & Country Rank (SJR), Egypt ranks 37th globally in scientific output, showing its potential but also highlighting the obstacles it faces in scientific and technological advancement.

2. Academia-Industry Collaboration: Egypt ranks 135th out of 142 countries in terms of collaboration between academia and industry. This reflects a significant disconnect between academic research and practical applications in industry.

3. Research and Development (R&D): According to the World Bank, Egypt's expenditure on R&D as a percentage of its GDP is approximately 0.71% (2020), which is considerably lower compared to other developing nations. Countries like South Korea invests more than 4% of their GDP in R&D.

4. Education System:

- Education spending: Egypt allocates approximately 3.8% of its GDP to education, which is below the global average of 4.5-5%. This impacts infrastructure, learning environments, and access to modern equipment.
- Classroom overcrowding: The student-to-teacher ratio in some schools exceeds 40 students per class, making it difficult to provide personalized attention or modern, technology-driven teaching methods.

5. Brain Drain: Approximately 45% of Egyptian scientists and researchers who study or work abroad do not return, further weakening the country's domestic R&D capabilities.

Global Comparisons:

- Countries like China and India have seen substantial improvements in their global rankings due to significant investments in R&D and education. Both countries are now ranked within the top 10 in the SJR rankings.

- Academia-industry collaboration in countries like Germany and Sweden ranks in the top 10 globally, showing how integrated research and innovation efforts lead to tangible economic benefits.

The Path Forward for Egypt:

To improve the scientific and technological environment, Egypt needs to:

- Increase R&D investment to match global standards, aiming for 1.5%-2% of GDP.
- Strengthen academia-industry collaboration through government incentives, partnerships, and research commercialization.

- Reform the education system to include more active, technology-based learning methods, with a focus on STEM (science, technology, engineering, and mathematics) subjects.

- Create programs to retain Egyptian scientists and encourage students who study abroad to return and contribute to domestic research.

By addressing these key areas, Egypt can better harness its potential in science and technology, which are fundamental for long-term growth and development.

3- Reduce and adapt to the effects of climate change:

Climate Change: Impact and Solutions in Egypt

Climate change manifests through rising temperatures, extreme weather events, sea-level rise, and shifting precipitation patterns. These changes disrupt ecosystems, agriculture, water resources, and infrastructure, posing significant risks to global economies. The Intergovernmental Panel on Climate Change (IPCC) estimates that the global economy could suffer losses of up to 3% of GDP by 2030 due to climate impacts, with regional variations in GDP loss as outlined in Table (2).

Table (1): shows an overview about the GDP loss due to the rise in temperatures.

	Temperature rise scenario, by mid-century			
	Well-below 2°C increase	2.0°C increase	2.6°C increase	3.2°C increase
	Paris target	*The likely range of global temperature gains*		*Severe case*
Simulating for economic loss impacts from rising temperatures in % GDP, relative to a world without climate change (0°C)				
World	−4.2%	−11.0%	−13.9%	−18.1%
OECD	−3.1%	−7.6%	−8.1%	−10.6%
North America	−3.1%	−6.9%	−7.4%	−9.5%
South America	−4.1%	−10.8%	−13.0%	−17.0%
Europe	−2.8%	−7.7%	−8.0%	−10.5%
Middle East & Africa	−4.7%	−14.0%	−21.5%	−27.6%
Asia	−5.5%	−14.9%	−20.4%	−26.5%
Advanced Asia	−3.3%	−9.5%	−11.7%	−15.4%
ASEAN	−4.2%	−17.0%	−29.0%	−37.4%
Oceania	−4.3%	−11.2%	−12.3%	−16.3%

Egypt’s Economy: The Impact of Climate Change Across Key Sectors

Egypt, located in a climate-vulnerable region, faces numerous challenges from rising temperatures, water scarcity, and sea-level rise. These changes have serious repercussions for Egypt’s economy, particularly in sectors like agriculture, water management, tourism, and public health.

1. Water Scarcity and Agriculture

- Nile Dependency: The Nile River provides over 97% of Egypt's freshwater, making it essential for agriculture and daily livelihoods.

- Water Availability: Shifts in rainfall and temperature patterns affect water supply, causing significant agricultural losses. According to the 2024 projections, agricultural production in Egypt could decline by up to 10% if these trends continue.

- Economic Impact: By 2060, climate-related disruptions to water availability and agriculture could result in a GDP loss of between 2% and 6%.

2. Coastal Vulnerability and Tourism

- Sea-Level Rise: Coastal cities like Alexandria are highly vulnerable to rising sea levels. Estimates show that 8% of Alexandria’s land area could be submerged by 2050 without intervention.

- Tourism: Egypt's tourism industry, a key contributor to the national economy, is also at risk. Coastal erosion and extreme weather events are damaging tourism infrastructure, with potential economic losses estimated at $5 billion annually by 2040 if climate change impacts are not addressed.

3. Health and Urban Population Growth

- Heat Stress: Rising temperatures are exacerbating public health issues, contributing to heat stress, cardiovascular diseases, and respiratory illnesses from increased air pollution.

- Urban Growth: Egypt's urban population is expected to grow by an additional 41.4 million people by 2050. As urban areas expand, climate vulnerabilities such as heatwaves and water shortages will disproportionately affect vulnerable communities, increasing poverty rates and social inequality.

Egypt’s Climate Solutions: Adaptation and Mitigation Strategies

In response to these challenges, Egypt has implemented several key initiatives aimed at reducing climate risks and fostering sustainable growth through clean technologies.

1. Renewable Energy – Benban Solar Park

- World-Leading Solar Farm: The Benban Solar Park in Aswan is one of the largest solar power installations in the world, covering 37 square kilometers.
- Energy Generation: The park generates 1.8 gigawatts of electricity, significantly reducing Egypt's dependence on fossil fuels.

- Global Climate Efforts: By expanding its renewable energy capacity, Egypt is contributing to global efforts to reduce greenhouse gas emissions and limit global temperature rise.

2. Water Management – Toshka Project

- Efficient Water Use: The Toshka Project is an innovative solution to Egypt's water scarcity, involving smart water management systems to divert water from Lake Nasser to the Toshka Depression in the Western Desert.

- Sustainable Agriculture: Using advanced irrigation systems, the project optimizes water use for agriculture, enhancing food security and climate resilience.

3. Coastal Protection – New Alamein City

- Green Urban Planning: New Alamein City, located along the Mediterranean coast, has been designed with climate resilience in mind. The city incorporates green infrastructure, such as mangroves and natural buffers, to protect against sea-level rise and storm surges.

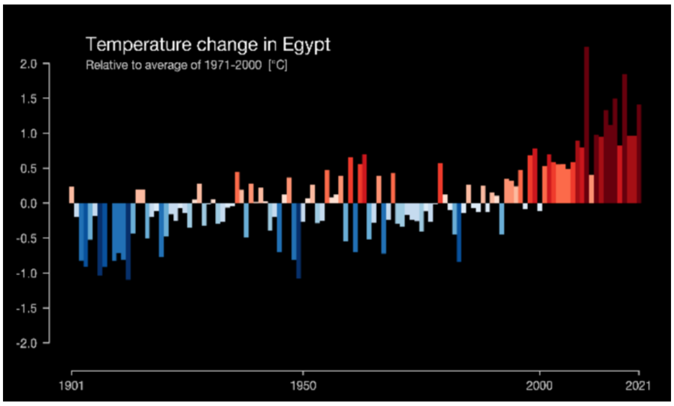
- Sustainable Development: This forward-thinking urban model prioritizes coastal defenses and integrates environmental sustainability, setting a standard for future urban developments in Egypt.

Alsp, Egypt's National Climate Change Strategy 2050 has been updated in 2024 to include more ambitious goals. The strategy now aims to increase renewable energy's share to 50% of the country's total energy mix by 2035, up from an initial target of 42%. Additionally, the country has committed to reducing greenhouse gas emissions by 15% by 2030 compared to 2015 levels, an update from previous targets due to increased international support and technological advancements.

By focusing on renewable energy, water management, and coastal protection, Egypt is taking proactive steps to mitigate climate risks, protect its economy, and foster sustainable development. These initiatives are crucial for safeguarding Egypt's future in the face of growing climate threats.

Problem to Be Solved.

Fig (1): The effect of climate change on the change of temperature in Egypt



Despite only about 0.6% of the world's yearly greenhouse gas emissions coming from Egypt, the country is Africa's second-largest gas producer and supplies nearly a third of the continent's gas, with production expected to further increase significantly in the upcoming years, both for domestic use and export to the EU. Low-income populations, both in Egypt and around the world, are disproportionately affected by several climate change-related risks. Egypt is particularly exposed to the effects of global warming with projected increases in heat waves, dust storms, storms along the Mediterranean coast, and extreme weather events. Stronger warming has been documented over the past 30 years, with average annual temperatures increasing by 0.53 degrees Celsius per decade as shown in Figure (X). The country's climate risks are and will impact today's younger generations. Crucially, the awareness of the importance of climate change action both domestically and at the global level is fast increasing in Egypt. The country is at a turning point in its commitment and action to tackle the consequences of climate change. In the 2030 Vision and Sustainable Development Strategy, Egypt has also made commitments to integrate climate change into national development policies and to green its budget across sectors progressively.

Egypt’s economic conditions make the problem even more difficult. For Egypt, agriculture is economically vital. Approximately 55 percent of the country’s labor force is involved in various agricultural activities, and in 2019 the sector formally employed 21 percent of Egyptian workers. The Ministry of Agriculture alone employs around 100,000 workers, making it the country’s second-largest employer. The agricultural sector generated about 11.3 percent of Egypt’s overall gross domestic product (GDP) in 2021. Moreover, its products satisfy around 30 percent of the county’s food requirements, making it a crucial source of both sustenance and income for countless Egyptians.

If the problem is solved:

- Egypt will be able to cope with the impacts of climate change on its water resources,
agriculture, coastal zones, and health sectors.

- Egypt will reduce its greenhouse gas emissions and contribute to the global efforts to mitigate climate change.

- Egypt will enhance its economic development and social welfare by investing in low-carbon
and climate-resilient technologies and practices.
- Egypt will strengthen its regional and international cooperation on climate action and share
its experiences and lessons learned with other countries.

If the problem is not solved:

- Egypt will face increasing water scarcity, droughts, floods, and salinization that will threaten
its food security and the livelihoods of millions of people.

- Egypt will lose its valuable coastal assets and infrastructure due to sea level rise and coastal erosion, affecting its tourism, fisheries, and urban development.

- Egypt will suffer from more heat waves, dust storms, and vector-borne diseases that will
harm its public health and well-being.

- Egypt will miss the opportunities to diversify its economy and create green jobs by relying
on fossil fuels and inefficient technologies.

Research

Topics Searched Related to the Problems:

- Climate challenges in Egypt.
- Air pollution issues.
- Water pollution and its environmental impact.
- Soil pollution and its impact on agriculture.
- Environmental monitoring gaps.

- Energy consumption in monitoring systems.
- Sensor accuracy challenges.
- Lack of weather data for remote and rural areas in Egypt.
- Seasonal weather variations and extreme temperature shifts.
- Current methods of environmental monitoring in Egypt.
- Power constraints for remote weather stations.
- Effects of pollution on human health and agriculture.

Topics Searched Related to the Possible Solutions:

- Energy-efficient monitoring systems.
- Kalman filters for improving sensor accuracy.
- Machine learning for environmental predictions.
- Solar-powered IoT solutions.
- Wireless data transmission in remote monitoring.
- Renewable energy solutions for powering weather stations.
- Low-cost sensor systems for large-scale deployment.
- Open-source weather data platforms.
- Predictive modeling for climate monitoring.
- Sustainable solutions for long-term environmental tracking.
- Smart technologies for real-time weather monitoring.
- Use of alternative energy sources to power remote sensors.

Sources: For resources, that we focused on there are cited websites such as gov, org, and edu. Also, scientific books or worksheets, and finally college professor.

Other Solutions Already Tried

European Centre for Medium-Range Weather Forecasts (ECMWF)

The European Centre for Medium-Range Weather Forecasts (ECMWF) is an intergovernmental organization based in Reading, United Kingdom. It is dedicated to providing numerical weather predictions and climate information services to its Member States and other partners.

ECMWF's primary focus is on producing accurate and reliable weather forecasts for the medium range, typically 1 to 15 days ahead. The Centre uses advanced numerical weather prediction models to simulate the atmosphere's behavior and generate forecasts. These models are based on complex mathematical equations that take into account various factors such as temperature, pressure, wind, humidity, and solar radiation.

Figure 3:

(ECMWF) picture

In addition to its medium-range forecasts, ECMWF also produces seasonal forecasts up to 6 months ahead and climate predictions for longer time periods. These forecasts are used by a wide range of organizations and individuals, including national meteorological services, government agencies, businesses, and researchers.

ECMWF is a leading international center for weather forecasting and climate research. Its work has contributed significantly to our understanding of the Earth's climate system and has helped to improve weather forecasts and climate predictions worldwide.

Advantages

- **Global Coverage:** ECMWF provides global weather forecasts, covering a wide range of regions and climates.
- **Accuracy:** The Centre's numerical weather prediction models are among the most accurate in the world.
- **Research and Development:** ECMWF conducts ongoing research to improve forecasting methods and develop new products.
- **Data Services:** Provides access to a vast database of meteorological data and information.
- **International Collaboration:** ECMWF is an international organization that fosters collaboration among its Member States.

Disadvantages

- **Computational Demands:** Running complex NWP models requires significant computational resources, which can be expensive.
- **Limited Accuracy:** While ECMWF's forecasts are generally accurate, there are limitations, especially for long-range predictions and extreme weather events.
- **Data Availability:** The quality and availability of observational data can impact forecast accuracy, especially in remote regions.
- **Climate Change:** Climate change is making weather prediction more challenging, as it can alter long-term patterns and increase the frequency of extreme events.

Specific Projects and Initiatives

ECMWF has been involved in numerous projects and initiatives related to weather forecasting and climate prediction. Some notable examples include:

- **Earth System Model:** ECMWF has developed an Earth System Model, which integrates atmospheric, oceanic, land surface, and cryosphere components to provide a more comprehensive understanding of the Earth's climate system.
- **Seasonal Forecasting:** ECMWF has made significant advancements in seasonal forecasting, providing predictions up to 6 months in advance.
- **Extreme Weather Events:** The Centre has focused on improving predictions of extreme weather events, such as heatwaves, droughts, and heavy rainfall.
- **Data Assimilation:** ECMWF has developed advanced data assimilation techniques to combine observations from various sources with NWP models.
- **Machine Learning:** The Centre has been exploring the use of machine learning to improve forecast accuracy and develop new products.

Overall, the European Centre for Medium-Range Weather Forecasts plays a crucial role in providing accurate and reliable weather forecasts and climate information. Its research and development efforts contribute to advancing the field of meteorology and improving our understanding of the Earth's climate system.

National Oceanic and Atmospheric Administration (NOAA) Weather Prediction Center

The National Oceanic and Atmospheric Administration (NOAA) Weather Prediction Center (WPC) is a division of the National Weather Service within the NOAA in the United States. It is responsible for providing national weather forecasts and warnings. The WPC uses a variety of advanced technologies to produce its forecasts, including numerical weather prediction models, satellite data, radar data, ensemble forecasting, and machine learning. The WPC's forecasts are used by a wide range of organizations and individuals, including emergency management agencies, media outlets, businesses, and individuals.
One of the WPC's most important functions is to issue severe weather warnings. These warnings include:

- **Thunderstorm Warnings:** Alerts to severe thunderstorms with damaging winds, hail, and lightning.
- **Tornado Warnings:** Alerts to the possibility of a tornado.
- **Hurricane Warnings:** Alerts to the approach of a hurricane.
- **Winter Storm Warnings:** Alerts to severe winter weather conditions such as heavy snow, ice, and high winds.

The WPC also provides forecasts for a variety of other weather phenomena, including:

- **Tropical Weather:** Forecasts for tropical cyclones, such as hurricanes and typhoons.

- **Hydrological Events:** Forecasts for flooding, drought, and other hydrological events.

- **Climate Prediction:** Long-term climate outlooks and predictions.
  The WPC's forecasts are used by a wide range of organizations and individuals. Emergency management agencies use the WPC's forecasts to plan for and respond to severe weather events. Media outlets use the WPC's forecasts to report on weather conditions and to provide weather forecasts to their viewers and listeners. Businesses use the WPC's forecasts to make decisions about operations and planning. Individuals use the WPC's forecasts to make personal decisions about travel, outdoor activities, and safety.
  The WPC is a vital part of the NOAA's mission to provide weather forecasts and warnings that protect lives and property. The WPC's forecasts are used by millions of people every day, and they play a critical role in helping people to stay safe during severe weather events.

Key functions and responsibilities:

- **National Weather Forecasts:** Issues national weather forecasts, including daily weather briefings, severe weather outlooks, and extended forecasts.

- **Severe Weather Warnings:** Issues severe weather warnings for phenomena such as thunderstorms, tornadoes, hurricanes, and winter storms.
- **Tropical Weather Forecasts:** Monitors and forecasts tropical cyclones, including hurricanes and typhoons.
- **Hydrological Forecasts:** Provides forecasts of river flooding, drought, and other hydrological events.
- **Climate Prediction:** Issues long-term climate outlooks and predictions.
  The WPC uses a variety of advanced technologies to produce its forecasts, including:
- **Numerical Weather Prediction (NWP) Models:** Employs state-of-the-art NWP models to simulate the atmosphere's behavior.
- **Satellite Data:** Leverages satellite imagery to monitor weather systems and track storms.
- **Radar Data:** Uses radar observations to detect precipitation and track storm movement.
- **Ensemble Forecasting:** Runs multiple NWP models with slightly different initial conditions to quantify uncertainty.
- **Machine Learning:** Applies machine learning techniques to analyze large datasets and improve forecast accuracy.
  The WPC's forecasts are used by a wide range of organizations and individuals, including:
- **Emergency Management Agencies:** For planning and response to severe weather events.
- **Media Outlets:** For reporting on weather conditions and forecasts.
- **Businesses:** For making decisions related to operations and planning.
- **Individuals:** For making personal decisions about travel, outdoor activities, and safety.

The National Oceanic and Atmospheric Administration (NOAA) Weather Prediction Center (WPC) provides a valuable service to the United States by providing national weather forecasts and warnings. However, like any forecasting system, the WPC has both advantages and disadvantages.

Advantages:

- **Accuracy:** The WPC's forecasts are generally accurate, especially for short-term forecasts.
- **Timeliness:** The WPC issues forecasts in a timely manner, allowing people to prepare for severe weather events.
- **Reliability:** The WPC has a long history of providing reliable forecasts.
- **Coverage:** The WPC provides forecasts for the entire United States, including both urban and rural areas.
- **Variety of Forecasts:** The WPC provides a variety of forecasts, including severe weather warnings, tropical weather forecasts, hydrological forecasts, and climate predictions.
  Disadvantages:
- **Limited Accuracy:** While the WPC's forecasts are generally accurate, there are limitations, especially for long-range forecasts and extreme weather events.
- **Uncertainty:** Weather forecasting is inherently uncertain, and the WPC's forecasts are subject to error.
- **Data Limitations:** The quality and availability of data can impact forecast accuracy, especially in remote areas.
- **Climate Change:** Climate change is making weather forecasting more challenging, as it can alter long-term patterns and increase the frequency of extreme events.

The Australian Bureau of Meteorology (BOM)

The Australian Bureau of Meteorology (BOM) is a government agency responsible for providing meteorological services to Australia. It offers a wide range of services, including national and regional weather forecasts, severe weather warnings, climate monitoring, hydrology, oceanography, and emergency services support. The BOM uses advanced technologies such as numerical weather prediction models, satellite data, radar data, oceanographic buoys, and climate models to produce its forecasts and services. These forecasts and services are used by government agencies, businesses, and individuals for decision-making, planning, and safety. The BOM plays a vital role in Australia's national infrastructure, providing essential information and services to the Australian public. The Australian Bureau of Meteorology (BOM) relies on a combination of advanced technologies and methodologies to produce its weather and climate forecasts and services. Here's a breakdown of the key mechanisms:

Data Collection and Processing:

- **Weather Stations:** A vast network of weather stations across Australia collects data on temperature, precipitation, wind, humidity, and other meteorological parameters.
- **Radar Systems:** Doppler radar stations provide real-time information on precipitation intensity, storm movement, and wind shear.
- **Satellite Imagery:** Satellites offer a wide-area view of weather systems, cloud cover, and ocean conditions.

- **Oceanographic Buoys:** Buoys deployed in the ocean collect data on temperature, salinity, currents, and wave height.

Numerical Weather Prediction (NWP) Models:

- **Complex Simulations:** NWP models use mathematical equations to simulate the atmosphere's behavior, accounting for factors like temperature, pressure, wind, humidity, and radiation.
- **Ensemble Forecasting:** Multiple NWP models are run with slightly different initial conditions to assess forecast uncertainty.
- **Data Assimilation:** Observations are combined with model forecasts to improve accuracy.

Post-Processing and Analysis:

- **Interpretation:** Meteorologists analyze model output and observational data to produce forecasts and warnings.
- **Quality Control:** Forecasts are reviewed and validated to ensure accuracy and consistency.
- **Communication:** Forecasts are disseminated through various channels, including public warnings, media, and online platforms.

Climate Modeling:

- **Global Climate Models (GCMs):** GCMs simulate the Earth's climate system, including atmosphere, ocean, land, and ice.
- **Regional Climate Models (RCMs):** RCMs focus on specific regions, providing higher-resolution climate projections.
- **Scenario Analysis:** Different climate scenarios (e.g., business-as-usual, emissions reductions) are explored to assess potential future impacts.

The BOM uses a variety of advanced technologies to produce its forecasts and services, including:

- **Numerical Weather Prediction (NWP) Models:** Employs state-of-the-art NWP models to simulate the atmosphere's behavior.
- **Satellite Data:** Leverages satellite imagery to monitor weather systems and track storms.
- **Radar Data:** Uses radar observations to detect precipitation and track storm movement.
- **Oceanographic Buoys:** Deploys oceanographic buoys to collect data on ocean conditions.
- **Climate Models:** Uses climate models to simulate future climate scenarios.

The BOM is a vital part of Australia's national infrastructure, providing essential information and services to the Australian public. Its forecasts and services help to protect lives and property, and to support economic activity.

Specific Projects and Initiatives:

- **Severe Weather Forecasting:** The BOM has developed advanced forecasting systems for severe weather events, such as thunderstorms, tornadoes, and cyclones.
- **Climate Change Research:** The BOM conducts research on climate change and its impacts on Australia.
- **Hydrology and Water Resources:** The BOM provides hydrological information and forecasts to support water resource management and flood mitigation.
- **Marine Forecasting:** The BOM provides marine weather forecasts and oceanographic information to support maritime activities.

- **Emergency Services Support:** The BOM works closely with emergency services to provide weather information and support during severe weather events.

Advantages and Disadvantages:

Advantages:

- **National Coverage:** The BOM provides weather and climate services for the entire Australian continent.
- **Accuracy:** The BOM's forecasts are generally accurate, especially for short-term predictions.
- **Reliability:** The BOM has a long history of providing reliable weather and climate information.
- **Variety of Services:** The BOM offers a wide range of services, including weather forecasting, climate monitoring, hydrology, and oceanography.
- **Public Education:** The BOM conducts public education programs to raise awareness about weather and climate issues.

Disadvantages:

- **Limited Accuracy:** While the BOM's forecasts are generally accurate, there are limitations, especially for long-range predictions and extreme weather events.
- **Data Limitations:** The quality and availability of data can impact forecast accuracy, especially in remote areas.
- **Climate Change:** Climate change is making weather forecasting more challenging, as it can alter long-term patterns and increase the frequency of extreme events.

Overall, the Australian Bureau of Meteorology plays a crucial role in providing essential weather and climate information to Australia. Its forecasts and services help to protect lives and property, support economic activity, and inform decision-making on a range of issues.

2

Generating **and Defending a Solution**

Solution & Design Requirements

Real-time Monitoring and Energy Efficiency:

The system operates by taking real-time data from the sensors every 90 seconds, with two real data and one predicted, this system ensures the data is captured each 30 seconds.

This approach minimizes energy through the use of a machine learning model which based on the data imported every 30 seconds will predict data for another 30 seconds putting the sensors in sleep mode in increasing and decreasing the total energy consumption over the course of the 6-month.

System Readings Accuracy and Reliability:

Accuracy and reliability will be key features of the rain-predicting system. The system will apply the real-time data to the Kalman filter, an algorithm that improves accuracy by predicting system states and updating them with new measurements, minimizing noise and outliers, and increasing the accuracy of the machine learning model.

Integrated Dashboard:

Dashboard showing the real-time data of the three sensors while showing the predicted data of the machine learning model. All are shown through graphs and tables for continuous human monitoring for faults or errors in the system.

Selection of Solution

Egypt, a country prone to sudden and severe weather events, requires innovative solutions to improve weather prediction and early warning systems. To address this challenge, we propose the development of a sophisticated weather monitoring system equipped with advanced sensors and artificial intelligence.

By strategically deploying temperature, pressure, and humidity sensors across the country, we can collect real-time data on atmospheric conditions. To optimize energy consumption and extend battery life, an AI model will be implemented. This model will analyze the sensor readings from the previous three minutes to predict the values for the next three minutes, allowing the sensors to enter a sleep mode during this period. This predictive capability will significantly reduce energy consumption, enabling the system to operate autonomously for up to six months.

The collected data will be stored on Google Sheets, providing a secure and efficient storage solution with a capacity of 10 gigabytes. This ample storage capacity will ensure that the system can operate continuously for one year without encountering storage limitations. Additionally, Google Sheets offers several benefits, including: accessibility from anywhere with an internet connection, collaborative capabilities for efficient teamwork, version history for easy review and restoration, seamless integration with other Google Workspace tools, and robust security measures to protect sensitive data.

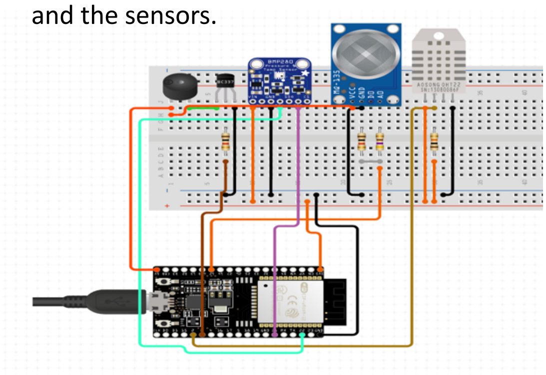
By combining advanced sensor technology, intelligent data analysis, and reliable data storage, this weather monitoring system will provide accurate and timely weather forecasts. This, in turn, will enable early warning systems to alert the public and authorities about impending storms, floods, and other extreme weather events, empowering them to take necessary precautions and minimize potential damage.

Selection of Prototype

To address the challenges of climate change in Egypt, we developed an innovative Smart Weather System that leverages an ESP32 microcontroller, multiple sensors, and advanced data processing techniques. By utilizing an AI model and a Kalman filter, this system provides accurate weather data and predictions, helping communities better understand and adapt to changing weather patterns.

ESP32 Microcontroller:

Fig(4): Schematic design of the microcontroller and the sensors.



The ESP32 serves as the core processing unit for our weather system. This microcontroller is cost-effective, energy-efficient, and equipped with Wi-Fi capabilities, enabling remote data transmission. It collects data from all connected sensors and performs initial processing, allowing real-time monitoring of environmental conditions. Its versatility and low power consumption make it ideal for sustainable, long-term climate monitoring solutions.

Sensors:

1. Temperature and Humidity Sensor:

The temperature and humidity sensors is essential for monitoring local weather conditions. It provides accurate data on the ambient temperature and moisture levels, which are crucial factors for understanding weather patterns and preparing for extreme conditions. This sensor data helps people make informed decisions, such as planning for heatwaves or humidity-related health precautions.

2. Air Quality Sensor:

To track pollution levels, we included an air quality sensor in the system. This sensor detects the concentration of harmful particles, such as PM2.5 and

PM10, which impact human health, particularly in urban areas. By providing data on air quality, the system empowers communities to take measures to reduce exposure to harmful pollutants, thus promoting better respiratory health.

3. Rainfall Sensor:

Monitoring precipitation is vital for agricultural planning and flood prevention. The rainfall sensor collects data on precipitation levels, which can be used to anticipate droughts or heavy rains. This helps farmers make informed decisions about crop irrigation and supports local authorities in preparing for potential flooding events.

Kalman Filter

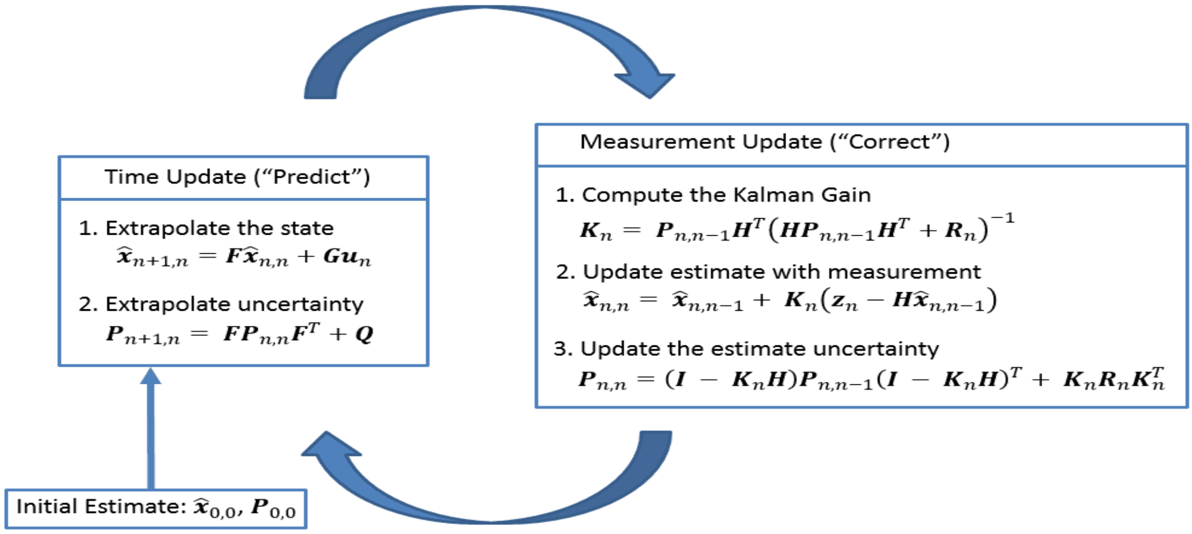
The Kalman Filter is a mathematical algorithm that provides estimates of unknown variables by combining a series of measurements observed over time, accounting for noise and inaccuracies. It's widely used in real-time applications such as signal processing, robotics, and navigation.

Key Principles and Rules:

- Prediction and Update Cycle: The Kalman Filter operates in two main steps: prediction and update.
  1. Prediction Step: It predicts the next state of the system based on the previous state and known system dynamics.
  2. Update Step: It updates the prediction based on new measurements, refining the state estimate by reducing uncertainty.
- State Estimation: The Kalman Filter maintains an estimate of the "state" of a system (e.g., position and velocity in navigation). The estimate is continuously updated based on a weighted average of predictions and new measurements.

- Measurement and Process Noise: The filter assumes that both the process (system) and measurements have Gaussian noise, characterized by a known variance. This allows it to weigh the measurements and predictions appropriately.

- Fig(5): Flow chart and Mathematical Formulas of Kalman Filter.



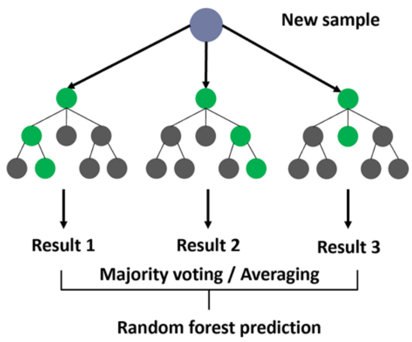
AI Model:

The Random Forest model is a supervised machine learning algorithm that uses an ensemble of decision trees for classification or regression tasks. It's popular in data science because of its flexibility and robustness.

Key Principles and Rules:

- Ensemble of Decision Trees: Random Forest creates multiple decision trees during training.

Fig(6): Schematic design for the architecture of the Random Forest AI model.



Each tree is trained on a random subset of the data, a technique known as bagging (Bootstrap Aggregation).

- Feature Randomness: At each split in a tree, only a random subset of features is considered for splitting, which helps reduce overfitting and makes the model more robust.
- Voting/Averaging for Prediction:
  - In classification, each tree votes on the class of an observation, and the class with the most votes becomes the model's final prediction.
  - In regression, each tree outputs a numerical value, and the final prediction is the average of these values across all trees.
- Out-of-Bag (OOB) Error: Random Forest uses samples not included in the bootstrap sample for testing each tree (called "out-of-bag" samples). This helps in estimating the model's accuracy without needing a separate test set.

Mathematical Formulation:

1. Training:
   - Randomly select samples from the training data with replacement (bagging).
   - For each sample, create a decision tree by choosing the best splits from a random subset of features at each node.
2. Prediction:
   - For classification, count the votes from each tree and select the class with the highest votes.
   - For regression, average the predictions from each tree.

   Rule Summary:

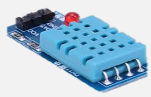
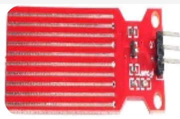
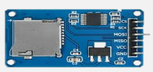
   Random Forest combines predictions from many trees to make a final decision, thus reducing overfitting and increasing generalization ability. The key rule is to take advantage of multiple decision trees, each trained on random samples and random subsets of features, to create a robust and stable model.

3

Constructing **and Testing a Prototype**

Materials

Total Cost | 870.00 L.E.

Item	Qua-ntity	Description	Usage	Cost	Source of Purchase	Picture
Arduino UNO	1	Microcontroller	Processing project data and code	350 L.E	Electronics store	
DHT-11	1	Temperature and humidity sensor	Temperature measurement	50 L.E	Electronics store	
BMP-280	1	Pressure, and temperature sensor	Monitors temperature and atmospheric pressure	100 L.E	Electronics store	
Battery	3	Power supplier	Supplies the project with energy	150 L.E	Electronics store	
Water Leakage Sensor	1	Water detection sensor	Detects water falls	90 L.E	RAM Electronics	
SD module	1	Data storage	Preserves collected data	120 L.E	RAM Electronics	

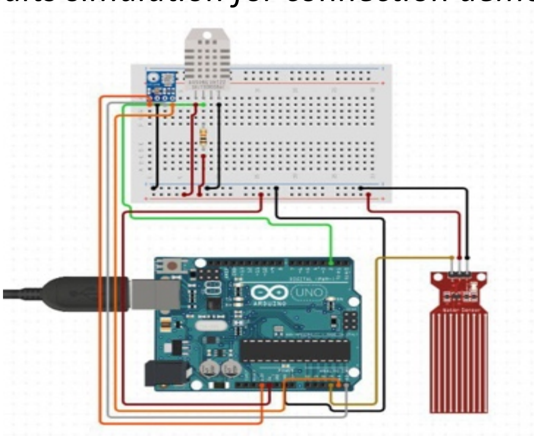
Methods

Prototype Development and Simulation

The prototype was initially simulated on **Circuito.io** to determine the necessary components and connection methods for building the weather station, as illustrated in Figure ().

Figure (7)

Prototype circuits simulation for connection demonstration.



Sensors for temperature, humidity, and barometric pressure were utilized to gather real-time data. Specifically:

DHT-11 (Temperature and Humidity Sensor): This sensor uses the one-wire protocol,

a time-based communication scheme that regulates communication between the microcontroller and the sensor.

BMP-280 (Barometric Pressure Sensor): This sensor employs the I2C communication

protocol, a master-slave communication method, allowing for clock-regulated data

transfer between the microcontroller and the sensor.

Improving Data Accuracy

After collecting data from the sensors, the measurements were further refined using

the **Kalman filter algorithm**. Data from the DHT-11 and BMP-280 sensors were Combined with readings from an accurate thermometer measuring temperature and humidity. The Kalman filter enhanced the precision of the data by reducing noise and ensuring high-quality inputs for further processing.

Preprocessing the Dataset for Machine Learning

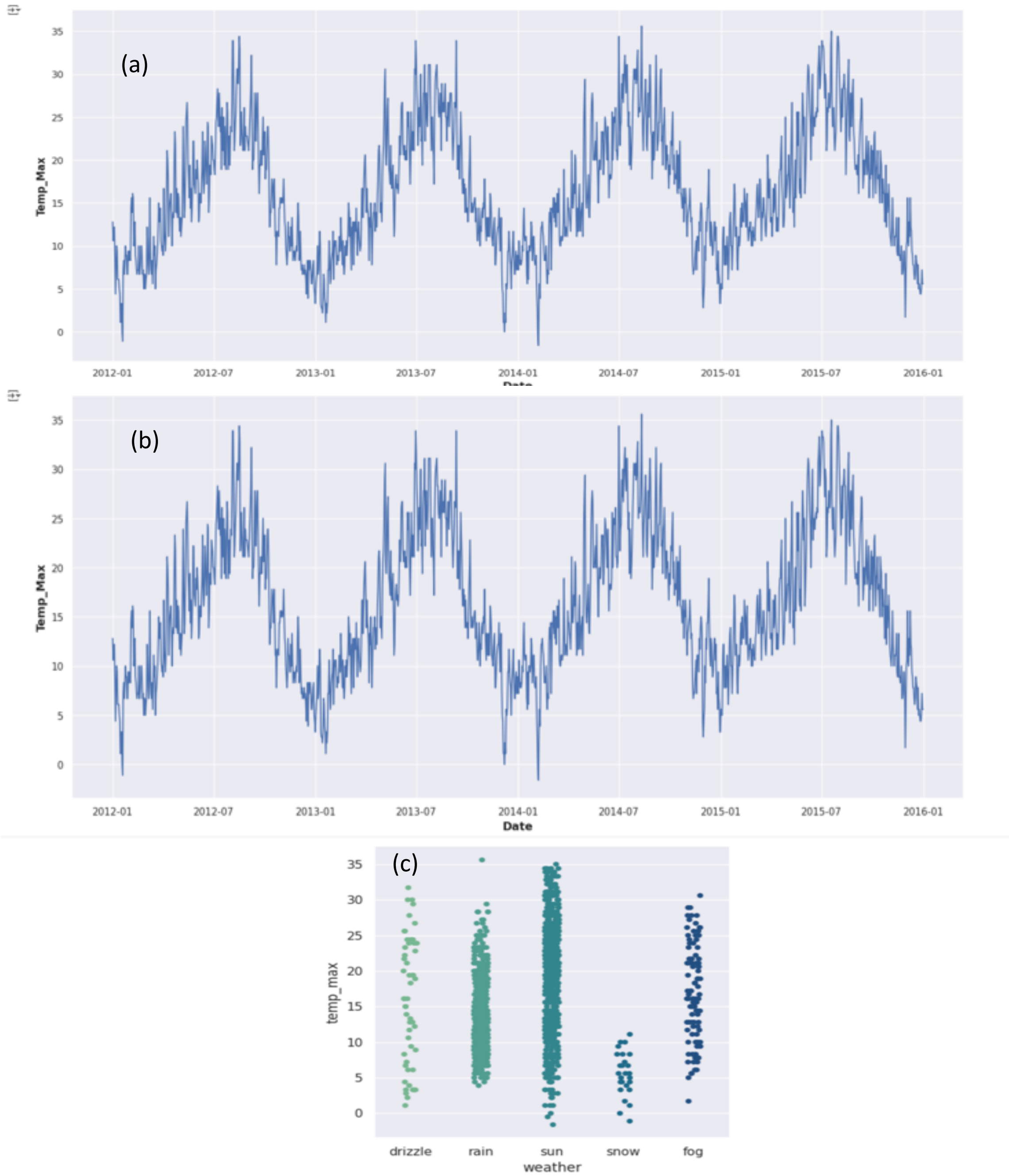
To train the machine learning model, a climate dataset containing Seattle's weather data from “**2012-01-01”** to “**2012-12-31”** was preprocessed using the **Pandas** and **NumPy** libraries. Key preprocessing steps included:

Handling Missing Data: Null values were identified and removed from the dataset.

Visualization: Data was plotted to ensure normal behavior and identify any anomalies, as shown in Figure (2).

Figure (8)

Data plotting for visualizing their behavior as shown in (a)-temperature maximum, (b)-temperature minimum, (c)-weather condition repetition across the year.



Training the Machine Learning Model

The **Naive Bayes classifier** from the **scikit-learn** library was used for training the machine learning model. The process involved:

Defining Variables:

A **training variable**: Containing the majority of the dataset, used for training the model.

A **target variable**: Containing the rest of the dataset, used for testing the model.

Model Training: The training variable was used to train the Naive Bayes classifier, which learned to predict the target variable.

Efficiency Analysis: The model's performance was evaluated using a **confusion matrix**, which broke down the number of true and false predictions.

Test Plan

A comprehensive test plan was conducted to evaluate the system's performance and energy efficiency:

Noise Reduction: The noise in the sensor data was analyzed before and after applying the **Kalman filter** in real time to quantify its impact on data quality.

Model Accuracy: The accuracy of the Naive Bayes model was measured using an **accuracy score**, which compared the number of correct predictions to the total predictions.

Energy Efficiency: The energy consumption of the prototype was assessed to quantify the reduction achieved after incorporating the Naive Bayes model.

Data Collection

Data accuracy enhancement through the Kalman filter and Naïve Bayes model accuracy

The real mean squared error was measured for performance indication of each set of data:

Figure (9)

Data collection from the BMP-280 pressure sensor before and after applying the Kalman filter:

(a)-pink before noise reduction by Kalman filter

(b)-blue after noise reduction by Kalman filter

Figure (10)

Data collection from the DHT-11 humidity sensor before and after applying the Kalman filter:

(a)-green- before noise reduction by Kalman filter

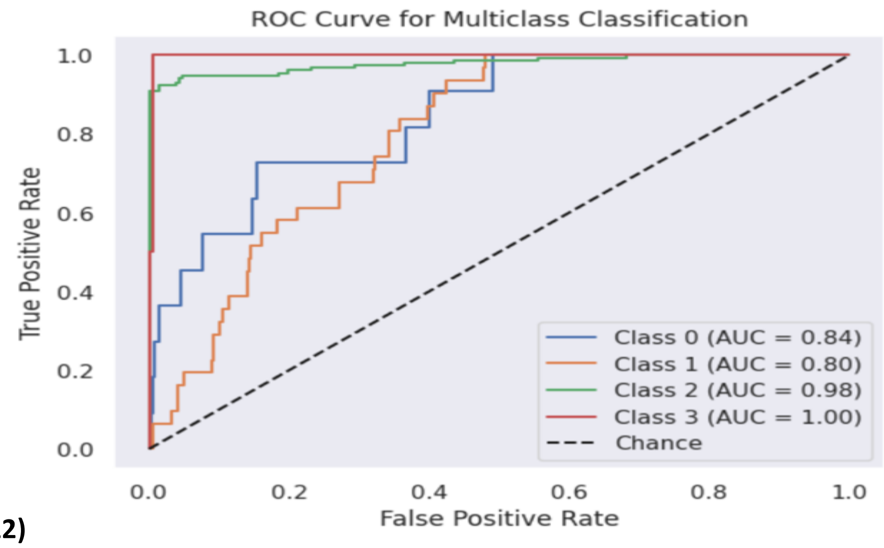
(b)-orange- after noise reduction by Kalman filter

The **real mean squared error** was measured for performance indication of each set of data:

RMSE temp: 1.8721, **RMSE humidity**: 4.002, **RMSE pressure**: 0.2430 **Naïve Bayes model accuracy**: 84.15%

Figure (11)

Confusion matrix that calculates the number of true values and false values



Figure(12)

AUC curve representing the relation between true and false positive values for each predicted class in the data: Drizzle, Foggy, Sunny, and Rainy.

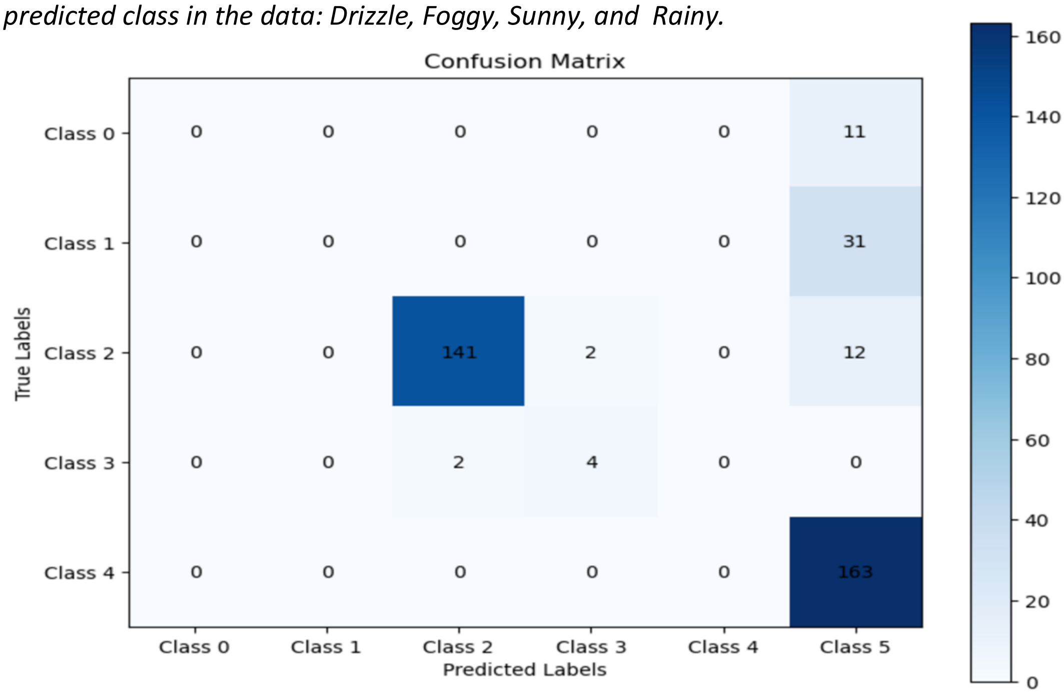


Table (2)

Energy consumption before and after applying ML and sleep mode.

Mode	Power consumption	Energy Consumption for 24 Hours
Normal Case	0.473W	11.352 Wh
Sleeping mode	0.473W (active), 0.00473W (sleep)	5.73276 Wh

4

Evaluation, **Reflection**, **Recommendations**

Analysis & Discussion

This section provides a detailed examination of how the proposed smart weather system effectively satisfied its two primary design requirements: **enhancing the accuracy of sensor readings** and **achieving high-accuracy weather predictions**.
Enhancing the accuracy of sensor readings:

Enhancing Accuracy of Sensor Data

Achieving the first design requirement was realized through the implementation of the Kalman filter. The Kalman filter is a mathematical algorithm widely used for estimating the state of a dynamic system in the presence of noise. The Kalman filter operates as a recursive estimator, meaning it updates its predictions as new data becomes available. This process consists of two primary stages: the **prediction step** and the **update step** ***(Lacey, 2019)***. The predicted state estimate is given by the equation

$\hat{x}k|k-1 = A\hat{x}k-1|k-1 + Bu_k$

where $\hat{x}k|k-1$ *represents the predicted state at time* k*,* A *is the state transition matrix,* $\hat{x}k-1|k-1$ is the previous state, and Bu_k accounts for control inputs. Simultaneously, the algorithm predicts the error covariance matrix, which quantifies the uncertainty.

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q.$$

Here, $P_{k|k-1}$ is the predicted error covariance, and Q is the process noise covariance, which models the uncertainty introduced by the system dynamics. In the update step, the Kalman filter refines its prediction by incorporating new sensor measurements. The innovation, or residual, which is the difference between the predicted measurement and the actual measurement, is computed and used to correct the state estimate:

$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1}$$

$$\hat{x}k|k = \hat{x}k|k - 1 + K_k(z_k - H\hat{x}k|k - 1)$$

where K_k is the Kalman gain, $z_k$is the actual measurement, H is the measurement matrix, and R is the measurement noise covariance. The Kalman gain dynamically adjusts the weight given to the measurement versus the prediction, minimizing the overall estimation error. Finally, the error covariance matrix is updated to reflect the reduced uncertainty:

$$P_{k|k} = (I - K_kH)P_{k|k-1}.$$

To assess the efficacy of the Kalman filter in improving sensor accuracy, we compared the **Root Mean Square Error (RMSE)** of the raw sensor readings to the RMSE of the filtered readings. The RMSE is a widely used metric for evaluating the accuracy of numerical predictions and is

defined as: $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ ***(STAT 3.01)***

Table (3)

Comparison between Raw RMSE and Filtered RMSE

Sensor	Raw RMSE	Filtered RMSE
Pressure (BMP 280)	1.3660	0.2430
Temperature (DH11)	3.6708	1.8721
Humidity (DH11)	11.5634	4.0032

Table (3) demonstrated a marked reduction in RMSE across all three parameters—temperature, humidity, and pressure—when using the

Kalman filter. For instance, the raw pressure readings from the BMP280 sensor exhibited substantial noise ***(PHYS 3.04)***, but the filtered data closely aligned with the actual values, achieving significantly lower RMSE. These findings confirm that the Kalman filter effectively enhanced sensor accuracy, thereby fulfilling the first design requirement.

Predicting weather conditions with high accuracy:

It was realized using a Naive Bayes classifier. Naive Bayes is a probabilistic machine learning algorithm based on Bayes' theorem, assuming conditional independence between features ***(Webb, 2016)***. Despite its simplicity, it is highly effective for classification tasks with structured and well-separated data. Bayes' theorem is expressed as:

$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$ ***(MATH 2.03)***

where P(C|X) is the posterior probability of class C given the input features X, P(X|C) is the likelihood of observing X given C, P(C) is the prior probability of C, and P(X) is the probability of the input data. In our application, the Naive Bayes model was trained on sensor data refined by the Kalman filter. Input features included temperature, pressure, humidity, and precipitation measurements, while the output represented weather conditions such as "clear," "rainy," or "cloudy." The model predicts the most likely weather condition for a given set of input features by maximizing the posterior probability.

The metric is computed as the area under the ROC curve, where the true positive rate (TPR) is plotted against the false positive rate (FPR) at various threshold levels. Mathematically TPR and FPR are defined as:

$TPR = \frac{TP}{TP+FN}, FPR = \frac{FP}{FP+TN}$ ***(Hussan et al., 2022)***
where TP and FP represent true positives and false positives, respectively, while TN and FN represent true negatives and false negatives. The system achieved an accuracy of **84.1%**, indicating reliable predictions of weather conditions. The AUC scores obtained during model evaluation were consistently high, indicating strong predictive accuracy.

Energy Consumption Calculations:

During the **Normal Case**: The system runs continuously for 24 hours with full power consumption of **0.473W while d**uring the **Sleeping Mode**: The system alternates between 30 seconds active and 30 seconds sleep. Power consumption during sleep is **1%** of the active power reducing power consumption to 50% of the Normal case.

$$P = V \times I \text{ (PHYSICS 2.05)}$$

$$P_{total} = P_{Arduino} + P_{DHT11} + P_{BMP280} + P_{Water\ Ptotal}$$

Recommendation

From solution success factors, it must be applicable on a large scale and useful in real life. As each project has its limitations due to its own set of conditions, there are some recommendations for future development:

· **Use Localized Data**: Utilize a dataset specifically designed for Egypt, focusing on local climate and agricultural conditions for more accurate predictions.

· **Real-Time Data Monitoring**: Implement a system to collect and analyze weather data in real-time for up-to-date forecasting and better disaster preparedness.

· **Enhance the Machine Learning Model**: Optimize the Naive Bayes algorithm by including local geographical and historical climate data for more accurate predictions.

· **Integrate Renewable Energy Solutions**: Incorporate renewable energy sources like solar power to reduce reliance on traditional energy, aligning with Egypt's sustainability goals.

· **Expand Data Storage Capacity**: Ensure sufficient data storage infrastructure to handle long-term data collection, analysis, and historical trend tracking for improved decision-making.

Learning Outcomes

Chemistry:

CH.2.11 Adopt renewable alternatives, reduce emissions, and reduce reliance on crude oil, promoting environmental sustainability and mitigating climate change impacts.

Mechanics:

MECH.2.05 Helped to be able to apply the concept of power to analyze energy transfer in mechanical systems. To calculate the power of airflow of the fan.

Mathematics

MATH 2.03 We Studied the fundmental counting principles, Permutations, combinations and factorials to be used in probability-based solutions.

STAT 3.01 We learned about Standard deviation, Speirman and pearson correlations

MATH.1.11 Matrix and linear programming were essential for modelling variables and analyzing data.

MATH.1.12 Determinants were used to evaluate matrices.

MATH.2.09 Area under the curve was an essential concept to evaluate the data efficiency through their plots.

English

ENG.2.17 We learned how to design a system of commuication through studying basic elements of communication, transmition and mobile phones

Physics

PHY 2.05 We utilized the concepts of Power estimation for circuit analysis.

PHY 3.04 We learned how to design a system of commuication through studying basic elements of communication, transmition and mobile phones.

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